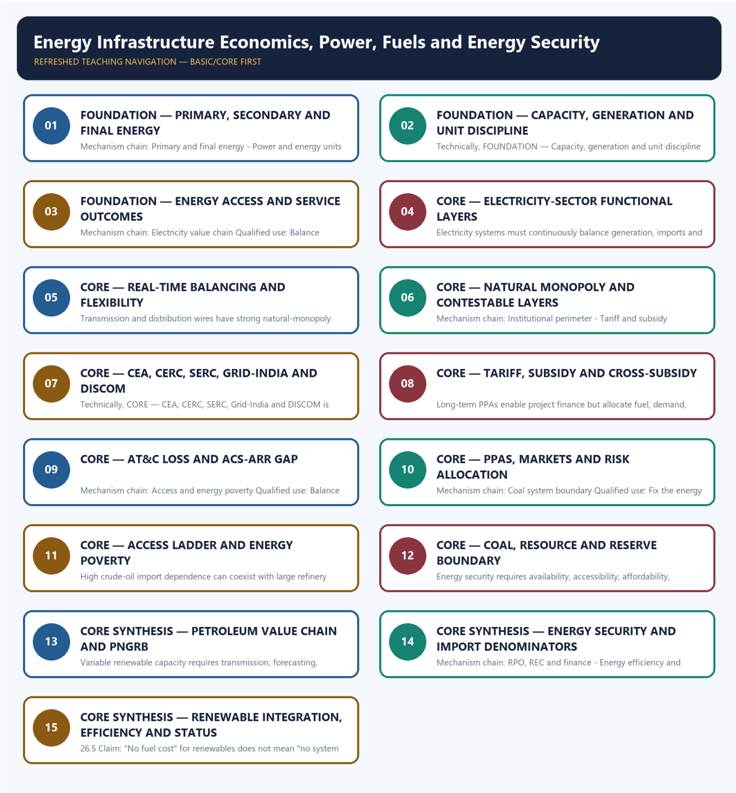
SOLVED PRACTICE WORKBOOK

Energy Infrastructure Economics, Power, Fuels and Energy Security - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Energy Infrastructure Economics, Power, Fuels and Energy Security - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Primary and final energy?

A. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. B. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. C. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. D. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable.

Answer: A. Explanation: Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q2. Which option preserves the accounting or regulatory boundary of Primary and final energy?

A. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. B. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. C. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. D. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response.

Answer: B. Explanation: Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q3. Which statement uses Primary and final energy without losing its vintage, basket or legal status?

A. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. B. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. C. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. D. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response.

Answer: C. Explanation: Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q4. Which option avoids the standard UPSC close-option trap about Primary and final energy?

A. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. B. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. C. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. D. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system.

Answer: D. Explanation: Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q5. Which statement correctly identifies Power and energy units?

A. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. B. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. C. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. D. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response.

Answer: A. Explanation: Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q6. Which option preserves the accounting or regulatory boundary of Power and energy units?

A. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. B. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. C. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. D. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response.

Answer: B. Explanation: Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q7. Which statement uses Power and energy units without losing its vintage, basket or legal status?

A. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. B. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. C. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. D. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply.

Answer: C. Explanation: Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q8. Which option avoids the standard UPSC close-option trap about Power and energy units?

A. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. B. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. C. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. D. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable.

Answer: D. Explanation: Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q9. Which statement correctly identifies Energy-service boundary?

A. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. B. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. C. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. D. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules.

Answer: A. Explanation: A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q10. Which option preserves the accounting or regulatory boundary of Energy-service boundary?

A. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. B. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. C. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. D. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response.

Answer: B. Explanation: A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q11. Which statement uses Energy-service boundary without losing its vintage, basket or legal status?

A. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. B. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. C. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. D. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply.

Answer: C. Explanation: A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q12. Which option avoids the standard UPSC close-option trap about Energy-service boundary?

A. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. B. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. C. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. D. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use.

Answer: D. Explanation: A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q13. Which statement correctly identifies Electricity value chain?

A. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. B. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. C. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. D. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply.

Answer: A. Explanation: Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q14. Which option preserves the accounting or regulatory boundary of Electricity value chain?

A. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. B. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. C. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. D. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules.

Answer: B. Explanation: Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q15. Which statement uses Electricity value chain without losing its vintage, basket or legal status?

A. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. B. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. C. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. D. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability.

Answer: C. Explanation: Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q16. Which option avoids the standard UPSC close-option trap about Electricity value chain?

A. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. B. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. C. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. D. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties.

Answer: D. Explanation: Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q17. Which statement correctly identifies Real-time balancing?

A. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. B. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. C. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. D. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences.

Answer: A. Explanation: Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q18. Which option preserves the accounting or regulatory boundary of Real-time balancing?

A. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. B. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. C. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. D. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences.

Answer: B. Explanation: Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility,

storage and demand response. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q19. Which statement uses Real-time balancing without losing its vintage, basket or legal status?

A. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. B. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. C. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. D. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences.

Answer: C. Explanation: Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q20. Which option avoids the standard UPSC close-option trap about Real-time balancing?

A. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. B. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. C. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. D. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response.

Answer: D. Explanation: Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q21. Which statement correctly identifies Natural monopoly and competition?

A. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. B. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. C. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. D. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences.

Answer: A. Explanation: Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q22. Which option preserves the accounting or regulatory boundary of Natural monopoly and competition?

A. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. B. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. C. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. D. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability.

Answer: B. Explanation: Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q23. Which statement uses Natural monopoly and competition without losing its vintage, basket or legal status?

A. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. B. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. C. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. D. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change.

Answer: C. Explanation: Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q24. Which option avoids the standard UPSC close-option trap about Natural monopoly and competition?

A. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. B. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. C. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. D. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules.

Answer: D. Explanation: Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q25. Which statement correctly identifies Institutional perimeter?

A. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. B. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. C. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. D. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences.

Answer: A. Explanation: CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q26. Which option preserves the accounting or regulatory boundary of Institutional perimeter?

A. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. B. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. C. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. D. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability.

Answer: B. Explanation: CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q27. Which statement uses Institutional perimeter without losing its vintage, basket or legal status?

A. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. B. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. C. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. D. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete.

Answer: C. Explanation: CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q28. Which option avoids the standard UPSC close-option trap about Institutional perimeter?

A. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. B. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. C. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. D. CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply.

Answer: D. Explanation: CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q29. Which statement correctly identifies Tariff and subsidy boundary?

A. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. B. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. C. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. D. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete.

Answer: A. Explanation: An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q30. Which option preserves the accounting or regulatory boundary of Tariff and subsidy boundary?

A. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. B. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. C. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. D. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete.

Answer: B. Explanation: An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q31. Which statement uses Tariff and subsidy boundary without losing its vintage, basket or legal status?

A. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. B. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. C. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. D. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete.

Answer: C. Explanation: An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q32. Which option avoids the standard UPSC close-option trap about Tariff and subsidy boundary?

A. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. B. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. C. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. D. An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences.

Answer: D. Explanation: An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q33. Which statement correctly identifies AT&C and ACS-ARR?

A. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. B. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. C. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. D. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change.

Answer: A. Explanation: AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q34. Which option preserves the accounting or regulatory boundary of AT&C and ACS-ARR?

A. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. B. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. C. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. D. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve.

Answer: B. Explanation: AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q35. Which statement uses AT&C and ACS-ARR without losing its vintage, basket or legal status?

A. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. B. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. C. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. D. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve.

Answer: C. Explanation: AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q36. Which option avoids the standard UPSC close-option trap about AT&C and ACS-ARR?

A. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. B. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. C. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. D. AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability.

Answer: D. Explanation: AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q37. Which statement correctly identifies PPA and procurement risk?

A. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. B. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. C. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. D. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete.

Answer: A. Explanation: Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q38. Which option preserves the accounting or regulatory boundary of PPA and procurement risk?

A. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. B. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. C. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. D. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas.

Answer: B. Explanation: Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q39. Which statement uses PPA and procurement risk without losing its vintage, basket or legal status?

A. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. B. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. C. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. D. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator.

Answer: C. Explanation: Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q40. Which option avoids the standard UPSC close-option trap about PPA and procurement risk?

A. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. B. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. C. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. D. Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change.

Answer: D. Explanation: Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q41. Which statement correctly identifies Access and energy poverty?

A. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. B. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. C. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. D. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas.

Answer: A. Explanation: Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q42. Which option preserves the accounting or regulatory boundary of Access and energy poverty?

A. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. B. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. C. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. D. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator.

Answer: B. Explanation: Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q43. Which statement uses Access and energy poverty without losing its vintage, basket or legal status?

A. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. B. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. C. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. D. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery.

Answer: C. Explanation: Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q44. Which option avoids the standard UPSC close-option trap about Access and energy poverty?

A. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. B. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. C. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. D. Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete.

Answer: D. Explanation: Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q45. Which statement correctly identifies Coal system boundary?

A. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. B. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. C. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. D. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas.

Answer: A. Explanation: Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q46. Which option preserves the accounting or regulatory boundary of Coal system boundary?

A. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. B. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. C. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. D. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator.

Answer: B. Explanation: Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q47. Which statement uses Coal system boundary without losing its vintage, basket or legal status?

A. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. B. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. C. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. D. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery.

Answer: C. Explanation: Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q48. Which option avoids the standard UPSC close-option trap about Coal system boundary?

A. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. B. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. C. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. D. Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve.

Answer: D. Explanation: Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q49. Which statement correctly identifies Petroleum chain and regulator?

A. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. B. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. C. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. D. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery.

Answer: A. Explanation: Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q50. Which option preserves the accounting or regulatory boundary of Petroleum chain and regulator?

A. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. B. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. C. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. D. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery.

Answer: B. Explanation: Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q51. Which statement uses Petroleum chain and regulator without losing its vintage, basket or legal status?

A. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. B. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. C. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. D. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost.

Answer: C. Explanation: Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q52. Which option avoids the standard UPSC close-option trap about Petroleum chain and regulator?

A. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. B. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. C. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. D. Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas.

Answer: D. Explanation: Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q53. Which statement correctly identifies Crude imports and product trade?

A. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. B. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. C. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. D. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery.

Answer: A. Explanation: High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q54. Which option preserves the accounting or regulatory boundary of Crude imports and product trade?

A. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. B. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. C. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. D. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation.

Answer: B. Explanation: High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q55. Which statement uses Crude imports and product trade without losing its vintage, basket or legal status?

A. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. B. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. C. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. D. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs.

Answer: C. Explanation: High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q56. Which option avoids the standard UPSC close-option trap about Crude imports and product trade?

A. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. B. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. C. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. D. High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator.

Answer: D. Explanation: High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q57. Which statement correctly identifies Energy-security dimensions?

A. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. B. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. C. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. D. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation.

Answer: A. Explanation: Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q58. Which option preserves the accounting or regulatory boundary of Energy-security dimensions?

A. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. B. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. C. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. D. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs.

Answer: B. Explanation: Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q59. Which statement uses Energy-security dimensions without losing its vintage, basket or legal status?

A. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. B. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. C. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. D. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body.

Answer: C. Explanation: Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q60. Which option avoids the standard UPSC close-option trap about Energy-security dimensions?

A. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. B. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. C. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. D. Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery.

Answer: D. Explanation: Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q61. Which statement correctly identifies Renewable integration?

A. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. B. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. C. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. D. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical.

Answer: A. Explanation: Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q62. Which option preserves the accounting or regulatory boundary of Renewable integration?

A. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. B. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. C. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. D. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body.

Answer: B. Explanation: Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q63. Which statement uses Renewable integration without losing its vintage, basket or legal status?

A. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. B. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. C. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. D. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body.

Answer: C. Explanation: Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q64. Which option avoids the standard UPSC close-option trap about Renewable integration?

A. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. B. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. C. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. D. Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost.

Answer: D. Explanation: Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q65. Which statement correctly identifies RPO, REC and finance?

A. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. B. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. C. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. D. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical.

Answer: A. Explanation: RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q66. Which option preserves the accounting or regulatory boundary of RPO, REC and finance?

A. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. B. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. C. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. D. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system.

Answer: B. Explanation: RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q67. Which statement uses RPO, REC and finance without losing its vintage, basket or legal status?

A. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. B. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. C. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. D. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system.

Answer: C. Explanation: RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q68. Which option avoids the standard UPSC close-option trap about RPO, REC and finance?

A. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. B. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. C. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. D. RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation.

Answer: D. Explanation: RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q69. Which statement correctly identifies Energy efficiency and rebound?

A. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. B. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. C. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. D. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system.

Answer: A. Explanation: Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q70. Which option preserves the accounting or regulatory boundary of Energy efficiency and rebound?

A. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. B. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. C. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. D. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body.

Answer: B. Explanation: Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q71. Which statement uses Energy efficiency and rebound without losing its vintage, basket or legal status?

A. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. B. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. C. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. D. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use.

Answer: C. Explanation: Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q72. Which option avoids the standard UPSC close-option trap about Energy efficiency and rebound?

A. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. B. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. C. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. D. Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical.

Answer: D. Explanation: Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q73. Which statement correctly identifies Fuel pricing and transition?

A. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. B. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. C. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. D. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system.

Answer: A. Explanation: Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q74. Which option preserves the accounting or regulatory boundary of Fuel pricing and transition?

A. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. B. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. C. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. D. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system.

Answer: B. Explanation: Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q75. Which statement uses Fuel pricing and transition without losing its vintage, basket or legal status?

A. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. B. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. C. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. D. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use.

Answer: C. Explanation: Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q76. Which option avoids the standard UPSC close-option trap about Fuel pricing and transition?

A. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. B. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. C. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. D. Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs.

Answer: D. Explanation: Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q77. Which statement correctly identifies Targets and legal status?

A. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. B. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. C. Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. D. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable.

Answer: A. Explanation: A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q78. Which option preserves the accounting or regulatory boundary of Targets and legal status?

A. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. B. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. C. Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. D. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use.

Answer: B. Explanation: A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q79. Which statement uses Targets and legal status without losing its vintage, basket or legal status?

A. A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. B. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. C. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. D. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response.

Answer: C. Explanation: A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

Q80. Which option avoids the standard UPSC close-option trap about Targets and legal status?

A. Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. B. Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. C. Transmission and distribution wires have strong natural-monopoly features, while generation, trading or retail arrangements may support competition under regulated access, settlement and reliability rules. D. A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body.

Answer: D. Explanation: A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

PYQS AND ANSWER PRACTICE

VERIFIED PYQ OWNERSHIP AUDIT

Audited ledgers route the 2018 energy-access, 2020 solar, 2021 Green Grid, 2022 renewable-target and 2025 clean-technology demands, plus objective concepts on PNGRB, coal institutions, solar regulation and ethanol feedstocks. The Basic/practice firewall carries them without inferring objective answers.

OWNER PYQ LEDGER EXTRACTS

20. PYQ closure

20.1 2018 GS-III - energy access and SDGs

Demand: “Access to affordable, reliable, sustainable and modern energy” as a development driver.

150-word architecture

1. Define access as connection + reliability + affordability + clean cooking.
2. Link to SDG 7 and transmission to health, education, livelihoods, gender and industry.
3. Progress: grid/household connections, clean cooking, efficiency and renewables.
4. Gaps: outages, quality, refill/consumption affordability, DISCOM viability, remote areas.
5. Way forward: reliable distribution, lifeline support, distributed energy, clean cooking,

productive-use finance and efficiency.

Conclusion: Count energy services and development outcomes, not only connected premises.

20.2 2020 GS-III - solar versus conventional energy

Demand: Benefits and government initiatives.

- economic: no fuel import, modularity, falling project costs, jobs, distributed access;
- environmental: low operating emissions and pollution;
- constraints: variability, land, grid, storage, materials, recycling and finance;
- initiatives: National Solar Mission, solar parks, rooftop programmes, PM-KUSUM, Green

Energy Corridors, auctions and RPO.

Judgement: Solar is central but requires grid, storage, manufacturing and land safeguards; it is not a standalone replacement for the whole power system.

20.3 2022 GS-III - renewable target and fossil-fuel subsidy shift

Answer in five steps:

1. Clarify the dated target and distinguish capacity from generation.
2. Explain security, pollution, climate and industrial gains.
3. Identify fossil support explicitly: consumer subsidy, producer support, tax concession,

below-cost finance or uncompensated externality are not identical.

4. Reform gradually through transparent accounting, carbon/efficiency signals and targeted

household support.

5. Invest savings in grids, storage, clean cooking, worker/region transition and innovation.

20.4 2021 GS-III - Green Grid Initiative and International Solar Alliance

Institutional facts

- FACT: The **International Solar Alliance (ISA)** was jointly launched by India and France at

COP21 in Paris in 2015; it is a treaty-based intergovernmental organisation headquartered in Gurugram.

- FACT: The **Green Grids Initiative-One Sun One World One Grid (GGI-OSOWOG)** was launched by

India and the United Kingdom at COP26 in Glasgow in 2021.

- WRONG: ISA and GGI-OSOWOG are not interchangeable names for one institution.

Purpose and economic logic

```
different time zones + diverse renewable profiles
 -> interconnected regional/transnational grids
 -> wider balancing area and solar-power exchange
 -> lower curtailment, reserve and storage pressure
 -> improved clean-energy access and system resilience
```

Constraints: cross-border transmission finance, technical standards, market settlement, sovereignty, cyber security, geopolitical trust and unequal cost-benefit distribution.

Answer judgement: Cross-border grids can complement domestic storage and flexibility, but cannot substitute for strong national grids, financially viable utilities or agreed governance.

20.5 2025 GS-III - energy independence through clean technology by 2047

The question has two demands; this owner closes the clean-technology half:

- efficiency and electrification;
- renewables plus transmission/storage/flexibility;
- nuclear and hydro where viable;
- domestic clean-equipment and critical-mineral strategy;
- green hydrogen/bioenergy for fit-for-purpose sectors;
- oil/gas diversification and strategic reserves during transition;
- DISCOM and power-market reform;
- recycling, R&D and just transition.

Biotechnology support includes advanced biofuels, compressed biogas, biomass conversion, enzymes/algae and waste-to-energy pathways; technical depth remains with S&T/Environment.

20.6 2025 GS-I - ecological and economic benefits of solar energy

Separate:

- **ecological:** lower operating emissions/air pollution, low operating water use relative

to thermal options, distributed deployment;

- **economic:** import substitution, price hedge, jobs/manufacturing, farm/rooftop income and

remote access;

- **qualification:** land, biodiversity, materials, recycling, variability and grid cost.

20.7 2025 Prelims - PNGRB and PM Surya Ghar

- PNGRB: midstream/downstream regulation within statute; crude/gas production is upstream.
- PM Surya Ghar: residential rooftop target plus training/capacity-building architecture;

distinguish target, sanction, installation and generation.

24. Ethanol, feedstock and biofuel policy trade-offs (declared PYQ demand)

24.1 India's Ethanol Blended Petrol (EBP) Programme - dated anchor

- CURRENT: India reached **20% ethanol blending in petrol (E20) during 2025**, reported by the

Ministry of Petroleum & Natural Gas as achieved **about five years ahead of the original 2030 target**, up from about 1.5% blending in 2014. Reported associated effects include foreign-exchange savings of roughly **Rs 1.36 lakh crore**, payments of roughly **Rs 1.96 lakh crore to distilleries** and **Rs 1.18 lakh crore to farmers** since the programme's scale-up, and an estimated reduction of about **698 lakh tonnes of CO_2** emissions. ANALYSIS: **Status caution:** these are government-reported programme totals as of the 2025/2026 official announcements; re-verify the exact figures and reporting date against the latest Ministry of Petroleum & Natural Gas/PIB release before quoting them as current, since cumulative totals are updated each supply year.

24.2 Feedstock comparison - why the 2025 Prelims question matters

Producer	Principal feedstock	Core economic logic
FACT: **Brazil**	Sugarcane (direct cane juice/molasses route)	High photosynthetic efficiency and an established cane-processing/sugar-mill base support large-scale, relatively low-cost ethanol production; flex-fuel vehicles absorb high blends.
FACT: **United States**	Maize/corn (starch route)	Vast domestic corn acreage and processing capacity support scale, but the starch-to-sugar conversion step is an additional cost layer compared with direct cane-juice fermentation.
FACT: **India**	Sugarcane juice/molasses **and** foodgrain (surplus rice/damaged grain, maize) - a dual-feedstock approach	Sugarcane supplies most current volume; grain-based capacity was added to accelerate the blending timeline and diversify feedstock risk.

24.3 Policy trade-offs this file must carry (not merely list feedstocks)

- **Food-versus-fuel trade-off:** diverting foodgrain/sugarcane to ethanol competes with food,

fodder and sugar-consumption uses; India's approach of prioritising sugarcane and surplus/ damaged grain (rather than diverting the entire foodgrain stock) is a deliberate mitigation, not a complete removal of the trade-off.

- **Water-intensity trade-off:** sugarcane is a water-intensive crop; expanding cane area for

ethanol can intensify the same groundwater/canal-water pressure already flagged for MSP- linked sugarcane cultivation (cross-link: Topic 28 §4.2-4.3 water-energy-crop nexus). Grain-based ethanol shifts (rather than removes) this pressure to grain-growing regions.

- **MSP/price-support interaction:** sugarcane's Fair and Remunerative Price and state advised

prices already create a minimum-price floor independent of ethanol demand; ethanol procurement adds a second demand channel that can support cane prices but also raise fiscal/pricing complexity if global sugar or ethanol prices move against the mill economics.

- **Import-substitution benefit versus engine/vehicle-compatibility cost:** higher blending

reduces crude-oil import dependence (linking to §13 energy-security), but requires E20-compatible engines, fuel-system materials and retail-infrastructure upgrades - a transition cost separate from the feedstock-supply question.

- **Diversification benefit versus single-crop dependence risk:** relying heavily on

sugarcane (as Brazil does) concentrates blending-programme risk on one crop's weather and price cycle; India's dual-feedstock design reduces this concentration risk but adds the food-security sensitivity noted above.

> MEMORY: **Prelims trap:** Brazil's ethanol economics and the US's ethanol economics rest on different feedstocks with different conversion chemistry (direct sugar fermentation versus starch hydrolysis); India's programme deliberately combines both feedstock families rather than replicating either country's single-feedstock model.

Recent PYQ Integration (2024-2025)

> **Status:** 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-PRELIMS-2024-2025.md`. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2025
- **Paper(s):** Prelims GS-I
- **Routed question demands:** 2

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2025	Prelims GS-I	60	Activities regulated by the Petroleum and Natural Gas Regulatory Board	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2025	Prelims GS-I	63	Ethanol producers Brazil and USA - feedstock comparison	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Activities regulated by the Petroleum and Natural Gas Regulatory Board
- Ethanol producers Brazil and USA - feedstock comparison

> This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

> **Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS3-GS4-2018-2023.md`, `_PYQ-ROUTING-PRELIMS-2018-2023.md`. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2020, 2021, 2022, 2023
- **Paper(s):** GS-III, Prelims GS-I
- **Routed question demands:** 8

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2018	GS-III	1	Energy access for Sustainable Development Goals in India	Comment · 10 marks · 150 words	Cross-routed to general-infrastructure and exam-complete Core energy-access owner	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2018	Prelims GS-I	67	Solar power production silicon wafers and tariff regulation India	Objective question; official key unavailable locally	Cross-routed to solar-technology and energy-regulation owners; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2019	Prelims GS-I	74	Petroleum Natural Gas Regulatory Board role and appeals	Objective question; official key unavailable locally	Regulatory, petroleum-sector and consolidated institutional owners; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2020	GS-III	16	Solar energy benefits versus conventional energy and government initiatives	Describe · 15 marks · 250 words	Cross-routed to renewable-technology and exam-complete Core energy-infrastructure owner	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2021	GS-III	6	Green Grid Initiative purpose at COP26 and ISA origin	Explain · 10 marks · 150 words	Cross-routed to climate/solar and exam-complete Core grid-integration owners	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	GS-III	12	Renewable energy 2030 target and shift from fossil fuel subsidies	Justify · 15 marks · 250 words	Cross-routed to climate-transition and exam-complete Core subsidy/energy-market owner	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2022	Prelims GS-I	72	Coal Controllers Organization statutory role and functions	Objective question; official key unavailable locally	Cross-routed to industrial-policy and coal-sector owners; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2023	Prelims GS-I	66	Coal-based thermal power plants India seawater water-stress	Objective question; official key unavailable locally	Cross-routed to spatial-resource and thermal-infrastructure owners; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Energy access for Sustainable Development Goals in India
- Solar power production silicon wafers and tariff regulation India
- Petroleum Natural Gas Regulatory Board role and appeals
- Solar energy benefits versus conventional energy and government initiatives
- Green Grid Initiative purpose at COP26 and ISA origin
- Renewable energy 2030 target and shift from fossil fuel subsidies
- Coal Controllers Organization statutory role and functions
- Coal-based thermal power plants India seawater water-stress

> The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

PYQ DEMAND CARD 1 - 2018 GS-III

Demand: Affordable, reliable, sustainable and modern energy as a development driver.

Status: Official-paper demand routed in the audited 2018-2023 GS-III ledger.

Model solution: Energy-service boundary: A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. **Access and energy poverty:** Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. **Energy-security dimensions:** Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. **Energy efficiency and rebound:** Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

PYQ DEMAND CARD 2 - 2022 GS-III

Demand: Renewable-energy target and the shift from fossil-fuel subsidies.

Status: Official-paper demand routed in the audited 2018-2023 GS-III ledger.

Model solution: Tariff and subsidy boundary: An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. **Energy-security dimensions:** Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. **Renewable integration:** Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. **Fuel pricing and transition:** Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. **Targets and legal status:** A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish installed power capacity, electricity generation and final energy service. Answer in about 150 words.

Model thesis: Claim: Primary and final energy. **Named evidence/example:** Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Power and energy units. **Named evidence/example:** Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Energy-service boundary. **Named evidence/example:** A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Claim -> named evidence -> analysis -> qualification:

- Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system.
- Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable.
- A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use.

Qualified conclusion: Claim: Primary and final energy. **Named evidence/example:** Primary energy exists in natural resources before conversion, while electricity and refined fuels are secondary carriers and final energy is what reaches end users; electricity is a power-system carrier, not the whole energy system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Power and energy units. **Named evidence/example:** Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Energy-service boundary. **Named evidence/example:** A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain why transmission ownership, system operation and regulation are distinct. Answer in about 150 words.

Model thesis: Claim: Electricity value chain. **Named evidence/example:** Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Real-time balancing. **Named evidence/example:** Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Institutional perimeter. **Named evidence/example:** CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Claim -> named evidence -> analysis -> qualification:

- Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties.
- Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response.
- CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply.

Qualified conclusion: Claim: Electricity value chain. **Named evidence/example:** Generation, transmission ownership, system operation, distribution, retail supply and market trading are distinct layers with different competition, coordination and regulatory properties. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Real-time balancing. **Named evidence/example:** Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Institutional perimeter. **Named evidence/example:** CEA provides statutory technical advice and planning, CERC regulates inter-state and central-sector matters within mandate, SERCs regulate intra-state and retail matters, Grid-India operates national and regional systems, and DISCOMs deliver and bill retail supply. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

ORIGINAL MAINS 3 - 15 MARKS

Question: Analyse DISCOM stress through tariff, subsidy, loss and procurement channels. Answer in about 250 words.

Model thesis: Claim: Tariff and subsidy boundary. **Named evidence/example:** An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AT&C and ACS-ARR. **Named evidence/example:** AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** PPA and procurement risk. **Named evidence/example:** Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Claim -> named evidence -> analysis -> qualification:

- An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences.
- AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability.
- Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change.

Qualified conclusion: Claim: Tariff and subsidy boundary. **Named evidence/example:** An observed below-cost tariff may be financed by explicit state subsidy, cross-subsidy, deferred regulatory recovery or DISCOM debt; these have different fiscal, distributional and investment consequences. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** AT&C and ACS-ARR. **Named evidence/example:** AT&C loss combines technical and commercial energy-recovery failures, while the ACS-ARR gap measures average cost versus realised revenue; reducing one does not guarantee financial viability. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** PPA and procurement risk. **Named evidence/example:** Long-term PPAs enable project finance but allocate fuel, demand, payment, curtailment, change-in-law and technology risks and can create stranded or inflexible cost when assumptions change. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

ORIGINAL MAINS 4 - 15 MARKS

Question: Examine India's petroleum value chain and PNGRB's regulatory perimeter. Answer in about 250 words.

Model thesis: Claim: Petroleum chain and regulator. **Named evidence/example:** Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Crude imports and product trade. **Named evidence/example:** High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Fuel pricing and transition. **Named evidence/example:** Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Claim -> named evidence -> analysis -> qualification:

- Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas.
- High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator.
- Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs.

Qualified conclusion: Claim: Petroleum chain and regulator. **Named evidence/example:** Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Crude imports and product trade. **Named evidence/example:** High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Fuel pricing and transition. **Named evidence/example:** Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate renewable integration through system value rather than capacity addition alone. Answer in about 300 words.

Model thesis: Claim: Power and energy units. **Named evidence/example:** Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Real-time balancing. **Named evidence/example:** Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Renewable integration. **Named evidence/example:** Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** RPO, REC and finance. **Named evidence/example:** RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Energy efficiency and rebound. **Named evidence/example:** Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Targets and legal status. **Named evidence/example:** A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Claim -> named evidence -> analysis -> qualification:

- Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable.
- Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response.
- Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost.
- RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation.

- Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical.
- A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body.

Qualified conclusion: Claim: Power and energy units. **Named evidence/example:** Installed capacity is rated power in MW or GW, while generation is energy produced over time in MWh, GWh or billion units; capacity share and generation share are not interchangeable. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Real-time balancing. **Named evidence/example:** Electricity systems must continuously balance generation, imports and discharge against demand, exports, charging and losses, giving economic value to forecasting, reserves, flexibility, storage and demand response. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Renewable integration. **Named evidence/example:** Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** RPO, REC and finance. **Named evidence/example:** RPO creates a renewable-procurement obligation, REC represents an eligible renewable attribute, and a finance company named REC Limited is a different institution; procurement rules do not by themselves create physical generation. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Energy efficiency and rebound. **Named evidence/example:** Efficiency supplies the same or better service with less energy input, but total energy use can still rise when output or use expands; energy conservation and efficiency are related but not identical. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Targets and legal status. **Named evidence/example:** A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design an energy-security strategy balancing affordability, access, resilience and transition. Answer in about 300 words.

Model thesis: Claim: Energy-service boundary. **Named evidence/example:** A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Access and energy poverty. **Named evidence/example:** Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Coal system boundary. **Named evidence/example:** Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Petroleum chain and regulator. **Named evidence/example:** Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Crude imports and product trade. **Named evidence/example:** High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Energy-security dimensions. **Named evidence/example:** Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Renewable integration. **Named evidence/example:** Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Fuel pricing and transition. **Named evidence/example:** Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Targets and legal status. **Named evidence/example:** A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

Claim -> named evidence -> analysis -> qualification:

- A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use.
- Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete.

- Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve.
- Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas.
- High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator.
- Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery.
- Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost.
- Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs.
- A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body.

Qualified conclusion: Claim: Energy-service boundary. **Named evidence/example:** A connection, sanctioned project or installed plant does not prove reliable, affordable, accessible or acceptable energy service at the point of use. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Access and energy poverty. **Named evidence/example:** Energy access is a ladder from network reach and connection to active metering, availability, reliability, affordability, productive use and clean cooking; connection counts alone are incomplete. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Coal system boundary. **Named evidence/example:** Thermal and coking coal have different uses and quality constraints, while mining, evacuation, plant stock, combustion, ash and closure form one infrastructure chain; geological resource is not the same as an economically recoverable reserve. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Petroleum chain and regulator. **Named evidence/example:** Upstream exploration and production differ from midstream transport and storage and downstream refining and marketing; PNGRB's statutory perimeter excludes production of crude oil and natural gas. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Crude imports and product trade. **Named evidence/example:** High crude-oil import dependence can coexist with large refinery capacity and petroleum-product exports; import dependence must state its commodity and denominator. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Energy-security dimensions. **Named evidence/example:** Energy security requires availability, accessibility, affordability, acceptability and resilience, including diversification of sources, routes and contracts, stocks, infrastructure, financial viability and shock recovery. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Renewable integration. **Named evidence/example:** Variable renewable capacity requires transmission, forecasting, flexible generation, storage, demand response, geographical diversity and market design; no fuel cost does not mean zero system cost. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Fuel pricing and transition. **Named evidence/example:** Retail fuel prices can reflect international prices, exchange rate, refining, marketing, freight, taxes and applicable support, while transition policy must separate consumer support, producer support and unpriced external costs. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Targets and legal status. **Named evidence/example:** A capacity target, planning estimate, scheme outlay, Bill, sanctioned project and achieved generation are separate status categories; current claims must retain the exact date, unit, legal stage and reporting body. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.